You can learn more about googleVis in the R interpreter, with the following:

- *help(googleVis)* which will give a brief description of the package and guide you to further information about it
- *demo(googleVis)* which will guide you through examples of charts included in the package, including gvisBubbleChart and gvisMotionChart
- *vignettes(googleVis)* which will give you an introduction to the package

Experimenting with stock market indices

The next step is to use your own data. An easy and interesting resource is to download historical indices data from the BSE (Bombay Stock Exchange) site. You can save the data as CSV files. The daily data will look something like what follows:

```
Date           Open      High      Low       Close
1-January-2016 26101.5   26197.27  26008.2   26160.9
4-January-2016 26116.52  26116.52  25596.57  25623.35
5-January-2016 25744.7   25766.76  25513.75  25580.34
```

You can merge and organise the data from the various CSV files in a spreadsheet so that it is suitable for importing in R. The YMD date format is very convenient as there is no ambiguity. You may want to add a column that shows the difference between the high and low values as an indicator of volatility. Suppose you want to look at the Sensex, MidCap, SmallCap and IT indices, the properly organised data will look like what follows:

```
Date       Index    Open    High     Low      Close    Hi-Lo
2016-02-29 Sensex   23238.5 23343.22 22494.61 23002    848.61
2016-02-29 Midcap   9584.31 9651.77  9389.35  9575.1   262.42
2016-02-29 Smallcap 9567.63 9594.03  9399.43  9548.33  194.6
2016-02-29 IT       10488.4 10507.62 10044.59 10229.49 463.03
```

Save this sheet in CSV format as *BSE-indices.csv*. This file can be easily imported as a data set in R, though the date field needs to be explicitly identified.

```
> D=read.csv("BSE-indices.csv", header=TRUE,colClasses=c
(Date="Date"))
> M <- gvisMotionChart(D,idvar="Index",timevar="Date",
sizevar="Hi.Lo")
> plot(M)
```

Notice that the 'Hi-Lo' becomes 'Hi.Lo' in R. This value is used as the default value for the size of the bubble in the above chart. The chart will open in a browser window and needs Flash Player. Figure 1 shows the initial image.

The *Open* column is used as the x-axis and the next column, *High*, is used as the y-axis. However, the columns used for the x and y axes can be changed in the chart. The colours for various indices can also be changed. If you

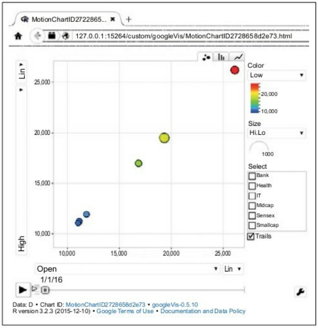

Figure 1: Initial display of a chart

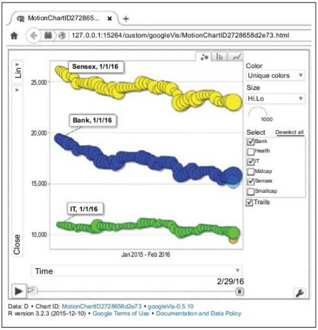

Figure 2: A chart showing some trails as the bubbles move in time